

SECTION 3

Copilot Across All Engineering Teams

Beyond Software — AI for Every Engineering Discipline



CAE



Design



Homologation



ADAS



SW Dev



Tools

Every team has unique workflows where Copilot and AI assistants can save hours per week

AI-Assisted CAE Workflows



Pre/Post Processing Scripts

Generate Python/TCL scripts for meshing automation, boundary conditions, and result extraction in HyperMesh, ANSA, or LS-PrePost



Solver Input Deck Review

Describe your LS-DYNA, Nastran, or Abaqus input deck — AI identifies keyword errors, missing contacts, or unit inconsistencies



Correlation Report Drafting

Generate test vs. simulation correlation reports with structure, metrics, deviation analysis, and improvement recommendations



Material Card Generation

Describe material behavior → AI drafts MAT keywords for LS-DYNA (*MAT_024, *MAT_058) or Nastran MAT cards



Result Interpretation Help

Share stress/strain descriptions → Get failure mode explanations, stress concentration causes, and design improvements



DOE & Optimization Setup

Design of experiments: parameter ranges, sampling strategies, response surface methodology guidance for optimization

CAE Pro Tip: Describe your simulation setup in plain text — AI helps debug solver errors without seeing proprietary models

CAE Prompt Templates

Debug Solver Error



"My LS-DYNA R14 crash simulation terminates at t=15ms with 'negative volume in element 34521'. The part is a B-pillar with shell elements (ELFORM=16), min element size 3mm. Material is *MAT_024 (DP600). What causes this and how do I fix it?"

Draft Correlation Report



"Draft a test-simulation correlation report for frontal NCAP. Test peak: 42g at 35ms. Sim: 39g at 33ms. Intrusion test: 85mm, sim: 78mm. Include methodology, comparison, root cause, improvement plan."

Generate Meshing Script



"Write a Python script for HyperMesh that: 1) Imports STEP file, 2) Creates component collectors by part name, 3) Applies 2D automesh target 5mm, 4) Checks element quality (min angle >30°, aspect ratio <5), 5) Exports as LS-DYNA keyword file."

Material Card Help



"Create LS-DYNA *MAT_058 card for CFRP composite. Fiber 0/90, E1=130 GPa, E2=10 GPa, strength 1500 MPa. Include failure criteria (DFAILM, DFAILS) and strain rate effects. Explain each parameter."

AI-Assisted Design Workflows



Design Review Documentation

Draft DR checklists, review minutes, and action items. Summarize design changes for gate reviews and cross-functional sign-offs



Tolerance Analysis & GD&T

Explain GD&T symbols, tolerance stack-up concepts, draft tolerance analysis reports. Datum references and MMC/LMC modifiers



BOM & Change Management

Draft ECRs, summarize BOM differences between versions, and generate impact analysis templates for design changes



Supplier Spec Clarification

Draft technical queries about component specs, material certifications, drawing clarifications. Bridge design intent and supplier language



DFMEA Assistance

Generate failure modes, effects, and causes for components. Suggest detection methods and recommended actions from function descriptions



CAD Scripting & Automation

Generate CATIA macros (VBA/CATScript), NX journaling scripts, or Creo Toolkit code for parameter updates and batch exports

Design Pro Tip: Use AI to draft DFMEA content — your experience validates severity, occurrence, and detection ratings

AI-Assisted Homologation Workflows



Regulation Interpretation

Explain complex ECE/FMVSS/GB clauses in plain language. Compare requirements across markets (EU, US, Japan, India, China)



Test Report Drafting

Generate type approval report structures, fill boilerplate sections, create compliance matrices mapping regs to test results



Regulation Change Tracking

Summarize changes between regulation versions (e.g., R155 rev2 vs rev3), identify impacted systems, draft gap analysis docs



CoP (Conformity of Production)

Draft CoP audit prep documents, sampling plans, and statistical analysis templates for production conformity verification



Technical Service Communication

Draft submissions to type approval authorities. Cover letters, clarification responses, and exemption requests



Multi-Market Compliance Matrix

Build comparison tables across regulations (lighting, emissions, safety) for variant management and platform strategies

Homologation Pro Tip: AI knows public regulation text — compare ECE R48 vs FMVSS 108 requirements, then verify against official gazette

AI-Assisted ADAS Development



Scenario Generation for Testing

Generate OpenSCENARIO test scenarios. Describe edge cases for AEB, LKA, ACC — AI suggests parameter variations and corner cases



Sensor Fusion Debugging

Describe fusion pipeline issues (camera+radar+lidar). Debug association logic, coordinate transforms, and tracking filter tuning



SOTIF & Safety (ISO 21448)

Draft SOTIF analysis templates, identify triggering conditions and functional insufficiencies, create V&V strategy documents



KPI Calculation & Analysis

Generate Python scripts for ADAS KPIs: TTC, time headway, lateral offset, false positive/negative rates from logged data



Euro NCAP Protocol Understanding

Explain Euro NCAP 2026 test protocols, scoring mechanisms, and color coding. Compare protocol changes between years



Data Pipeline & Labeling

Write Python for processing driving logs: parsing ROS bags, extracting CAN signals, converting labels (COCO, KITTI, nuScenes)

ADAS Pro Tip: AI excels at edge-case scenarios — describe the ODD and let AI suggest adversarial conditions you might miss

AI-Assisted Software Engineering



Code Generation & Review

Generate C/C++ embedded code, Python test scripts, and shell automation. Reviews for memory leaks, buffer overflows, CERT C



Requirements Engineering

Write EARS-pattern requirements, check ambiguity/testability, generate review checklists, and draft DOORS import formats



ASPICE Process Documentation

Draft work products for SWE.1–SWE.6: requirements, architecture, design, unit tests, integration test plans, traceability



Diagnostic Implementation (UDS)

Generate CDD content, DTC definitions, UDS service implementations (\$22, \$2E, \$31). ISO 14229 services and NRC handling



CI/CD & DevOps Automation

Create Jenkins pipelines, Docker configs, GitLab CI YAML for embedded builds. Debug build errors and linker script issues



Communication & Test Scripts

Debug CAN/LIN/Ethernet issues, generate CAPL scripts for CANoe, write protocol conformance test specifications

SW Dev Pro Tip: GitHub Copilot in VS Code auto-completes embedded C — use it alongside M365 Copilot for documentation tasks

AI-Assisted Tool Administration



Tool Configuration & Scripting

Generate config scripts for DOORS, Polarion, Jira, Teamcenter. Automate provisioning, workflow customization, report generation



License & Infrastructure Mgmt

Draft license usage analysis, capacity planning docs, and tool ROI assessments. Scripts to monitor FlexLM/RLM license servers



Integration & API Development

Generate REST API integration code (Jira ↔ Polarion, DOORS ↔ Excel). Write data migration scripts and ETL pipeline configs



User Training & Documentation

Draft tool user guides, quick-start tutorials, FAQ pages, and training presentations for common engineering tool tasks



Troubleshooting & Support

Generate diagnostic scripts for tool issues, automate log collection, draft known-issue databases. Parse error logs and fix



Process Automation (RPA)

Design Power Automate flows, write PowerShell for batch operations, create macros for repetitive data entry across tools

Tools Pro Tip: Use AI to generate REST API calls — describe the tool and desired action, AI writes the Python/PowerShell code

GitHub Copilot: Your Coding Co-Pilot



What GitHub Copilot Does

Real-time code completion as you type in VS Code, Visual Studio, JetBrains

Generates entire functions from comments or function signatures

Writes unit tests, docstrings, and boilerplate automatically

Copilot Chat: ask questions about your codebase in natural language

Supports C, C++, Python, MATLAB, CAPL, Shell, YAML, and 20+ languages



Automotive Engineering Use Cases

● **Embedded C:** MISRA-aware code suggestions for ECU software

● **CAPL Scripts:** Auto-complete CANoe test scripts from comments

● **Python Testing:** Generate pytest fixtures & HIL test automation

● **CI/CD YAML:** Complete Jenkins/GitLab pipeline configurations

● **ARXML Gen:** Help generate AUTOSAR description file patterns

M365 Copilot vs GitHub Copilot: M365 for docs & communication. GitHub Copilot for actual coding. Both = maximum productivity

AI Prompts for Every Engineering Discipline

Team	Use Case	Example Prompt
CAE	Solver Debug	"My Nastran SOL 101 fails with FATAL: SINGULAR MATRIX. Model has RBE2/RBE3 elements. What causes this?"
Design	DFMEA Draft	"List potential failure modes for a fuel tank filler neck. Include effects, causes, severity (1-10), detection methods."
Homol.	Reg Compare	"Compare ECE R13H and FMVSS 135 braking requirements. Table: parameter, ECE value, FMVSS value, key differences."
ADAS	Test Scenarios	"Generate 10 edge-case AEB scenarios for urban intersection. Include pedestrian, occlusion, lighting variations."
SW Dev	CAPL Script	"Write CAPL for CANoe: monitor CAN 0x100, check DLC=8, log if 'EngineSpeed' exceeds 6500 RPM."
Tools	API Integration	"Write Python to read Jira issues labeled 'safety-critical' via REST API, export to Excel: key, summary, status."

Universal Rule: Replace [brackets] with your specifics. Include version numbers, tool names, and constraints for best results

A Day with Copilot: Every Engineer's Story



CAE

Teams: "Summarize crash sim review" → Word: "Draft correlation report" → Excel: "Analyze energy balance trends"



Design

Outlook: "Draft ECR notification" → Word: "Generate DFMEA entries for new bracket" → PPT: "Create DR gate presentation"



Homol.

Edge: "Summarize R155 amendments" → Word: "Draft compliance gap analysis" → Outlook: "Email technical service update"



ADAS

Chat: "Generate 20 cut-in scenarios for ACC" → Excel: "Calculate false positive rate" → Word: "Draft SOTIF analysis"



SW Dev

VS Code: "Auto-complete CAPL script" → Word: "Draft ASPICE SWE.3 evidence" → Teams: "Summarize code review meeting"



Tools

Chat: "Write PowerShell to audit DOORS" → Word: "Draft tool migration guide" → Teams: "Summarize outage post-mortem"

Every team saves 5-10 hours/week on documentation, communication, and repetitive tasks



What NOT to Upload — Team-Specific Guide

Team	NEVER Upload	SAFE to Describe
CAE	FE models, mesh files, material test data, crash results	Generic solver errors, meshing strategies, material model keywords
Design	CAD files, drawings, BOM data, tolerance specs for parts	GD&T concepts, generic DFMEA patterns, ECR templates
Homol.	Type approval docs, test results, VIN data, CoC certs	Public regulation text, generic compliance matrices, templates
ADAS	Sensor data, driving logs, algorithm code, model weights	General ADAS concepts, OpenSCENARIO patterns, KPI definitions
SW Dev	Source code, ARXML, CAN DB (.dbc), calibration (.a2l/.hex)	Error messages, design patterns, AUTOSAR concepts, MISRA rules
Tools	Tool credentials, license keys, network configs, user data	Generic API patterns, scripting help, tool workflow concepts

Golden Rule: Describe the PATTERN, not the proprietary implementation. AI knows the technology — it doesn't need your specific data

Everything You Need to Remember



AI Science

LLMs predict tokens, not truth. Hallucinations are built-in. Math is text, not calculation. Context fades in the middle. Always verify.



Copilot Power

M365 Copilot = GPT-4 + YOUR data. Meetings, emails, docs — all AI-assisted. The "/" command is your secret weapon for context.



Prompt Engineering

CRISP framework: Context, Role, Intent, Scope, Preference. 5 min on prompting saves 30 min on iterations. Be specific, not vague.



Every Team Benefits

CAE, Design, Homologation, ADAS, SW Dev, Tools — all teams save 5-10 hrs/week on documentation, communication, and repetitive tasks.



Safety First

Describe patterns, never upload IP. AI helps you WRITE faster — YOU are responsible for CORRECTNESS. The assessor asks YOU, not AI.



Start Today

Try ONE technique this week. Create ONE reusable prompt. Track time saved. Share success. AI is a co-pilot — you're still the pilot.

"AI won't replace automotive engineers. But engineers who use AI will replace those who don't."

Start small. Stay curious. Share success. Protect Nissan IP always.

9+

Hours saved
per week

70%

Faster first
drafts

85%

Meeting notes
time saved

6

Teams
empowered